a STAGE 1 Technical scoreability

137 → **132** → **42**

strict genes → not-known-incompatible → structure-ready

97.91% row scoreability ≠ **3 balanced genes**

MutPred-PPI: 10,917/11,150 submitted rows scored

b STAGE 2 Phenotype-balanced representation

MutPred-PPI	9 with C4 ≥70%	→	**3 balanced**
SPPIDER-seq	127 with C4 ≥70%	→	**23 balanced**
X-PAIR	107 with C4 ≥70%	→	**15 balanced**

c STAGE 3 Incremental phenotype information

15 shared → 10 phenotype-ready → 547 residues

Median $\Delta R^2 = 0.0018$
95% CI includes zero

GLOBAL FAIL

Evaluation order, not causal flow. Failed gates remain frozen.